

Microstructure and sulfonation of dual monomer-grafted polypropylene nonwoven fabrics

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ABSTRACT: The sulfonation of styrenic polymers often needs harsh reaction conditions, such as concentrated sulfuric acid and elevated temperature. The awkward situation is handled by grafting copolymerization of polar and nonpolar monomers, and essentially the surface wettability between sulfonating agent and matrix polymer is circumvented in the study. Two pairs of dual monomers, glycidyl methacrylate (GMA)-acrylic acid (AA), styrene (St)-acrylic acid (AA), were grafted onto polypropylene nonwoven fabrics (PP) using thermal polymerization, and the resulted fabrics were further sulfonated using sodium sulfite and sulfuric acid, respectively. The microstructure of the modified fabrics was characterized by infrared spectroscopy and scanning electron microscopy. The ion exchange property was verified by copper (II) removal from aqueous solution, and its reutilization was carried out in electrochemical desorption process. The results show that hydrophilic and active monomer AA speeds grafting and sulfonation, but overload of AA will deteriorate copolymerization of GMA, and switch copolymerization chain component due to acylation with benzene ring. Although interfacial compatilizer promotes dual grafting, its overdosing may intensify microphase segregation of the grafting polymers owing to emulsification. The resultant granular polymers will be drained off during functionalization and reutilization processes. The ion exchange fabrics can be applied for treatment of metal ion wastewaters with unique electrochemical desorption feature. © 2019 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* **2019**, 136, 48300.

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INTRODUCTION

Recently, membrane separation techniques for dyeing and chemical wastewater treatments in environment engineering have drawn increasing attention. Polypropylene (PP) nonwoven fabric is a promising material for membrane separation due to many advantages like cheap, available, and durable. But its nonpolarity features strongly restrict their application in membrane separation technique. The grafting modification of polar monomers is an efficient method to solve this problem. However, PP easily undergoes chain scission in the course of melt grafting. A so-called comonomer concept thus has been developed to improve the melt grafting yield of maleic anhydride onto PP backbone chains.^{1,2} Nevertheless, the melt-grafting yield is generally low due to a short reaction time.

The grafting copolymerization of PP products, such as pellets, films, fibers, and fabrics, often takes place in the suspensions or the interfaces between its solid products and liquid monomers. In the situations, reaction temperature is much low, and reaction

time is quite a long duration, thus a high grafting yield can be anticipated rationally. The use of binary comonomer system is one of most viable pathway to enrich functionality and reactivity of PP substrate by introducing two different chemical nature groups.

The removal of various heavy metal ions from aqueous solutions was remarkably improved by grafting sodium styrene sulfonate (SSS) onto carboxymethyl cellulose gels because of better swelling capacity.³ The anticoagulant effect of poly(vinyl alcohol)-g-(acrylic acid (AA)-SSS) was found to be different from those of single monomer grafted versions but was much similar to that of heparin.⁴ Kulkarni *et al.* found that sulfonic and carboxylic bifunctional nut husk adsorbent demonstrated a better selective in comparison to monofunctional ion exchange resins IRC50 and IR120 for Pb(II) uptake over Cu(II), Zn(II), Cd(II), Mn(II), Fe(II), Co(II), and Ni(II) in an equimolar concentration in the binary metal system.⁵ Zu *et al.* prepared a bifunctional ion exchange battery separator membrane possessing faster alkali absorption and shorter total response and recovery times *via*

preirradiation graft copolymerization route.⁶ Totura insisted that the ion exchange resin of diphosphonic acid reacted completely with metal ions in solution, while not being consumed by alkaline earth metal elements, but the inclusion of sulfonic group in the resin would improve ion access to the diphosphonic sites and speed exchange rate.⁷ Proton exchange membranes for high temperature fuel cell was developed by grafting copolymerization of AA and hydroxyethyl methacrylate onto poly(ethylene-alt-tetrafluoroethylene) film in order to overcome a low binding capacity of AA with phosphoric acid.⁸ During electrodeionization process, carboxylic-type exchange resins have weak ion adsorption, strong desorption, and poor conductivity, *vice versa* for sulfonic-type exchange resins.⁹ Apparently the copolymerization of carboxylic and sulfonic monomers is good choice for cation exchange resin instead of mixing filling of the two type resins.

Styrene (St) is often grafted onto polymer fiber and film surfaces since the following-up functionalization can be realized on the benzene ring, however, sulfonation was implemented in caustic conditions. The sulfonation of polyethylene-g-glycidyl methacrylate (GMA) membrane even lasted for a few days.¹⁰ PP-g-St particles were sulfonated by being immersed in 98% H₂SO₄ in a weight-to-volume ratio of 1/5 at 98 °C for 2 h,¹¹ while the poly(vinylidene fluoride)-g- St membranes were functioned by being soaked in 98% H₂SO₄ at 70 °C for 8 h.¹² Nevertheless, sulfonation of St-hydroxyethyl acrylate-lauryl methacrylate) terpolymer took place at room temperature for a few hours.^{13,14} Benavides *et al.* prepared a novel proton conductivity membrane by copolymerizing St and AA and successively sulfonating in sulfuric acid (H₂SO₄, 98%) at equimolar ratio even for 1 h at ambient temperature.¹⁵

Binary monomer grafting presents quite different copolymerization behavior and properties. The effects of grafting parameters, such as irradiation dose, comonomer composition, and type of solvent, on the yields of grafting onto PP fiber were investigated, and those results clearly showed a decrease grafting with the increase of feed acrylonitrile.¹⁶ That grafting behavior was explained by precipitation grafting of acrylonitrile rich copolymers from solvent. The grafting yield of butyl methacrylate and acrylamide comonomers onto PP films increased with the amount of acrylamide increasing till a maximum value at the mass ratio of one,¹⁷ which was interpreted by high reactivity and polarity of acrylamide. Radiation grafting copolymerization of binary mixtures of AA and St onto PP films was investigated by El-Salmawi *et al.* They concluded that a highest grafting yield was obtained at a solvent composition of 20% water/80% methanol and a comonomer mixture of 20% AA/80% St, and their reactivity ratios of AA and St were found to be 0.45 and 1.3, respectively.¹⁸

To overcome the incompatibility between highly ionized sulfonic group and hydrophobic polyethylene backbone, Reddy *et al.* employed a two-step procedures, that is, preliminary grafting of AA chains to improve hydrophilicity of polyethylene, then successive graft polymerization of SSS after another irradiation.¹⁹ However, the graftability of SSS was finally circumvented in the presence of N-vinyl-2-pyrrolidone²⁰ and AA²¹ in a one-step process. Lee *et al.*²² reported that a highest value of grafting for AA

and SSS was about at their molar ratio of one. It was thought that overload of AA got reaction solution viscosity increased and eventually monomer diffusion decreased in AA-homopolymer segment, whereas overload of SSS would also decrease in grafting due to remaining hydrophobic surface.

The binary monomers of glycidyl acrylate and AA were photo-grafted onto high density polyethylene sheet to improve adhesion. That strong adhesion was considered to relate less branched graft chains.²³ The work does imply the two monomers really copolymerize into a same polymer chain. Sequential radiation grafting of N,N-dimethylaminoethyl methacrylate and N-isopropyl acrylamide made PP film surface to possess high affinity to diclofenac, while maintaining to regulate drug release and to avoid inflammatory response.²⁴ However, continual twice irradiation would deteriorate the substrate mechanical properties.

The grafting sites normally locate in amorphous region of semi-crystalline PP fibers and films, and the high crystallinity of PP fibers and films relatively restrain free-radical access. In fact, their SEM images demonstrated that the sequential graftings of AA and dimethyl aminoethyl methacrylate on PP film were sparsely scattered granules.²⁵ Similar morphological feature was also observed by Ghaffar *et al.*¹⁷ Granular structures were also found on the high-density polyethylene film surface grafted in different solvents, and each granule was thought to be a single graft chain.²⁶ In order to improve the distribution uniformity of grafting polymers on PP fiber surface, a combination of conformal grafting technique and high dosage of photo-initiator was conceived and implemented.²⁷

Poly(ethylene terephthalate) (PET) fibers modified by copolymerization grafting of GMA and methacrylic acid were used for removal of methylene blue (MB) in an aqueous solution. The surface charge nature of the grafted PET fibers can be easily switched by adjusting the pH values of dye aqueous solution based on the deprotonations of carboxylic and epoxy groups. Desorption of the MB-adsorbed fibers was conducted in the acetic acid/methanol mixture.²⁸ The binary monomers vinylimidazole and acrylonitrile-grafted polymer chains on the PP fibers showed considerable affinity towards heavy metal ions, and their affinity reached a maximum at the molar ratio of one, which was speculated to generate alternating copolymer chains and to coordinate maxim adsorption capacity.¹⁶

In this study, the main objective was to explore the feasibility to easily sulfonate the styrenic polymers at mild condition, and the resulted fabrics would find the application for dyeing and heavy metal ion wastewater separation and purification. Two pairs of dual monomers, GMA-AA and St-AA, were grafted onto PP fabrics by means of thermal polymerization, respectively. The ion exchange fabrics were further sulfonated by using sodium sulfite and concentrated sulfuric acid respectively. The microstructure, adsorption, and electrochemical desorption of the functioned fabrics were investigated.

EXPERIMENTAL

Materials

Spun-bonded PP nonwoven fabric (100 g/m²), Kunshan Sanyang Nonwoven Materials Co. Ltd, Jiangsu, China, heterogeneous

anion exchange membrane (made from LDPE mixed with ion exchange resin, thickness 0.48 mm, ion exchange capacity 1.8 mmol/g), Zhejiang Qianqiu Environmental Water Treatment Co., Ltd., Zhejiang, China. St, GMA, AA, divinylbenzene (DVB), pentaerythritol triacrylate (PETA), benzoyl peroxide (BPO), China Pharmaceutical Group Chemical Reagents Co., Ltd., are of analytical grade and were used without purification. Concentrated sulfuric acid, sodium sulfite, methanol, acetone, isopropanol (IPA), copper sulfate pentahydrate, sodium dodecyl benzene sulfonate (SDBS), China Pharmaceutical Group Chemical Reagents Co., Ltd., are of analytical grade. Rapeseed oil, Auchan Supermarket, Suzhou, Jiangsu, China.

Grafting Process

Preparation and Sulfonation of PP-g-(GMA-AA) Fabrics. First, the PP fabric was washed by acetone and deionized water in order to remove the oil agents on PP fabric and dried at 50 °C overnight. Then, the as-cleaned fabric was pretreated by glow-like discharging for 5 min at irradiation 20 mW/mm² in order to improve wettability toward grafting solution and generate free radicals and/or hydroperoxides but free from mechanical damage (handle type, CTP2000K Plasma Generator, Nanjing Suman Plasma Technology Co. Ltd., Jiangsu, China). Third, the discharged fabric was impregnated in the grafting monomer solution, monomers (AA, GMA), initiator BPO, crosslinker PETA, and rapeseed oil as an interfacial compatilizer, and the superfluous solution on fabric was removed with filter paper. Fourth, the loaded fabric was sandwiched with Teflon sheets and placed in the oven (DZF-6051, Shanghai Precision Scientific Instrument Co. Ltd, Shanghai, China) at 120 °C to keep polymerizing for 40 min. Finally, the grafted fabric was soaked in mixture solvent methanol/acetone at volume ratio of 1/1 for 5 h and further cleaned with deionized water to remove remnant monomer and homopolymer, and vacuum dried at 50 °C overnight. The grafting yield was calculated based on weight gain as following:

$$Gy = \frac{M_y - M_0}{M_0} \times 100 \quad (1)$$

where, G_y is grafting yield, M_0 and M_y are the fabric weight prior to and after grafting.

In addition, the other two series experiments were also done as follows. The first four graftings were made with the volume ratios of GMA to AA 3/0, 3/1, 3/2, and 3/3 (molar ratio 1.0/0, 1.44/1.0, 1.0/1.39, and 1.0/2.09) based on the optimal condition except for monomer ratio as fixed (abbreviated as optimal run). The resulted fabrics were denoted as PP-g-GMA, PP-g-(GMA3-AA1), PP-g-(GMA3-AA2), and PP-g-(GMA3-AA3). The other four grafting runs were carried out with the compositions: the volume ratio of monomers/PETA/compatilizer = 7/3/2, the monomers were AA, GMA, GMA7/AA3, and GMA3/AA 7(v/v) (abbreviated 7/3/2 run).

The ingredients of sulfonating solution for PP-g-(GMA-AA) were sodium hydroxide, sodium sulfite, isopropanol, and deionized water at the corresponding ratio of 0.2 g/20 g/30 mL/ 150 mL²⁹ (one sulfonation was run without sodium hydroxide). The sulfonating system was evenly stirred and filled into 250-mL Erlenmeyer flasks, respectively. One gram of the grafted fabrics

was ensured immersion in the flasks, and then the flasks were set into water bath oscillated at 80 °C for 5 h (HD500 water bath oscillator, Nantong Hongda Experimental Instruments Co. Ltd., Jiangsu, China). The sulfonated fabrics were cleaned by deionized water with ultrasonic cleaner (LC500, Shandong Jining Luchao Ultrasound Equipment Co. Ltd, Shangdong, China), and then dried in a vacuum at 50 °C overnight. The sulfonated PP-g-(GMA-AA) was abbreviated as S-PP-g-(GMA-AA).

Preparation and Sulfonation of PP-g-(St-AA) Fabrics. Forty milliliters of St, 10 mL of DVB, 2.5 g of BPO, and 4 g of SDBS were ground and mixed, and then a fixed volume of AA [volume ratio of St to AA was 3/0, 3/1, 3/2, and 3/3 (namely 1/0, 1.70/1.0, 1.0/1.18, and 1.0/1.77 [molar])] was further dosed and mixed evenly. Only 1 mL of rapeseed oil was added into the 3/2 recipe. Also another run with the volume ratio of St/AA at 3/0, 3/1, 3/2, and 3/3 was done but in the absence of SDBS and oil. The impregnated fabric was sandwiched with Teflon sheets and grafted in an oven stepwise (70 °C 30 min, 100 °C 20 min, and 130 °C 20 min). The grafted fabric was further immersed in 60 and 98% sulfuric acid at ambient temperature for 1 h, respectively, and then cleaned with deionized water, and dried in a vacuum at 50 °C overnight. The sulfonated fabrics were abbreviated as S-PP-g-(st-AA), S-PP-g-(st3-AA1), S-PP-g-(st3-AA2), and S-PP-g-(st3-AA2).

Copper(II) Adsorption

Adsorption of S-PP-g-GMA Fabrics. A total of 0.5 g of S-PP-g-GMA fabrics were immersed in 100 mL 50 mg/L as Cu²⁺ solution for 20 min at room temperature, and the copper (II) content of the remnant solution was sampled and tested. The copper removal was calculated based on Cu(II) concentration variation

Adsorption and Electrochemical Desorption of S-PP-g-(St-AA) Fabrics.

S-PP-g-(St-AA) fabrics with the diameter at 50 mm were cut and pretreated in 0.5 M NaOH aqueous solution for 24 h, and rinsed with deionized water, and then immersed in aqueous solution of copper (II) 1000 mg/L in a water bath at 40 °C for 8 h. Afterwards, the samples were rinsed with deionized water. The copper (II)-adsorbed fabric was leaned against the positive electrode clamped with plastic grid, while the heterogeneous anion exchange membrane was covered on the negative one. The electrolyte in electrolytic cell was copper sulfate at the initial concentration copper (II) 1000 mg/L. Electrochemical desorption was carried out at DC voltage 20 V (KA6002D DC power supply, Shenzhen Keruiyuan Technology Co., Ltd, Guangdong, China).

The adsorption capacity (A_c , %) was calculated by the ratio of weight reduction of copper ions in solution to the weight of the input adsorbent fabric.

$$A_c = \frac{(C_0 - C) \times V}{M_y} \times 100 \quad (2)$$

where, A_c is Cu(II) adsorption capacity, C_0 and C are Cu(II) concentrations at initial and 8 h after adsorption, respectively, V is Cu(II) solution volume.

The electrochemical desorption rate (E_d , %) was defined as the weight ratio of the copper ions weight gain in solution to the copper ions weight adsorbed on the fabric.

The electrochemical desorption current efficiency (C_e , %) was calculated based on the ratio of the molar electric charge of copper ions desorbed to the total electric charge transfer in solution.

$$C_e = \frac{z \times F \times n}{\int_0^t I dt} \times 100 \quad (3)$$

where, C_e is desorption current efficiency, n is molar number of Cu(II) desorbed into solution, z is ionic valence, and F is Faraday's constant 96 500 C/mol, I and t are the desorption electric current and time, respectively.

For the regeneration and repetitive utilization of the sulfonated fabric, the size and pretreatment of S-PP-g-(St3-AA2) fabric, as well as adsorption and electrochemical desorption operations, were same as mentioned above, and adsorption and electrochemical desorption operation was repeated three times.

Characterization

Morphology of Grafted Fabrics. The fabric surface morphology was observed by scanning electron microscopy (Hitachi, SU1510) after being gold sputtered. The elemental analysis of the functionalized fabrics was conducted by SEM-EDS. The sulfonation yield was calculated according to the elemental contents acquired by SEM-EDS. For the samples S-PP-g-GMA, all carbon atoms were assumed to come from PP, PP-g-GMA, and S-PP-g-GMA, and all oxygen atoms from PP-g-GMA and S-PP-g-GMA, and all sulfur atoms from S-PP-g-GMA. Sulfonation only was supposed to happen on epoxy groups.

For the samples S-PP-g-(GMA-AA), GMA fraction was assumed to be sulfonated completely, and the assumption was verified by comparing the calculated and experimental grafting yields. The molar grafting yield was defined as the molar ratio of monomer grafting to matrix PP. The conversion of the molar grafting into the weight grafting was realized by multiplying the molar mass ratio of grafting monomer to PP. Similarly, for the sample S-PP-g-(st-AA), St moiety in grafting chains was assumed to copolymerize with AA solely with no acylation reaction occurred. Also the grafted polystyrene was speculated to be sulfonated completely, and the molar grafting yield was thus calculated.

Microstructural Analysis. The infrared spectra of the pulverized fabrics were pressed into diskette with potassium bromide and recorded with Fourier infrared spectrometer Nicolet 5700 (Thermo Fisher Scientific Inc.) in a resolution of 2 cm^{-1} .

Copper Contents. The copper ion content in solution was measured by inductively coupled plasma atomic emission spectrometer, ICAP 6300 DUO (Thermo Fisher Scientific Inc.).

RESULTS AND DISCUSSION

Binary Monomers Grafting Modification and Sulfonation

As shown in Figure 1, the typical adsorption peaks for PP appeared at 1375, 997, and 967 cm^{-1} come from methyl group and chain configuration in the crystalline region. Generally, the

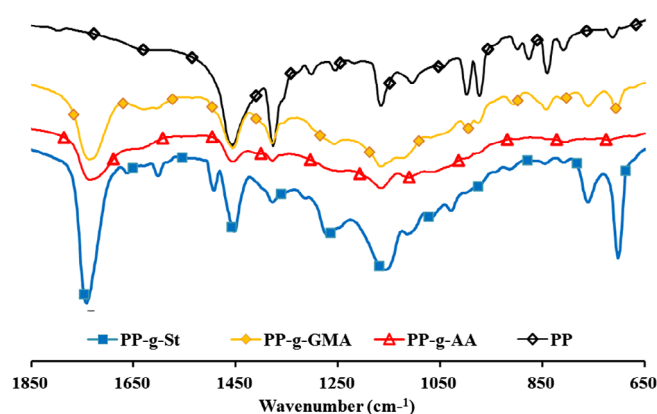


Figure 1. Infrared spectra of PP prior to and after grafting with three different monomers (7/3/2 run) (Note: PP-g-St means polypropylene fabric grafted with polystyrene. GMA and AA mean the monomers of glycidyl methacrylate and acrylic acid. 7/3/2 run indicates the volume ratio of monomer/PETA/compatilizer is equal to 7/3/2, and the monomers are AA, GMA, and St.) [Color figure can be viewed at wileyonlinelibrary.com]

reaction activity of AA and its esters can be much similar. However, the introduction of methyl group in GMA increases the hyperconjugation effect, and thus the grafting yield of GMA is decreased, which leads to incomplete coverage of GMA grafting on PP fiber, as evidenced the presence of the twin peaks at 997 and 967 cm^{-1} . The weak adsorption peaks due to carboxylic ionization of AA appear at 1550 and 1410 cm^{-1} . The broadening at low frequency side of adsorption peak approximately 1710 cm^{-1} from carbonyl group also arises in PP-g-AA. The relatively weak peak at 895 cm^{-1} can be identified and is assigned to the epoxy group of PP-g-GMA. The St grafting can be easily detected from the characteristic peaks at 700 cm^{-1} and almost twin peaks at 1499 and 1450 cm^{-1} . The peak at 1735 cm^{-1} emerged in all spectra of grafted PP, together with two weak peaks at 1245 and 1050 cm^{-1} , are assigned to ester group from crosslinker—PETA.

Orthogonal experimental is a design method for a set of multifactor and multilevel experiments. It is based on orthogonality to select some representative points from comprehensive runs. The representative points possess the characteristics of uniform dispersion, neatness, and comparability. It is an efficient, fast, and economical experimental design method. In this grafting copolymerization, the main factors include the monomer volume ratio, initiator, crosslinker, interfacial agent, and immersing time. The grafting was run according to the test schedule of $L_{16}(4^5)$, namely, 5 factors and 4 levels, totally 16 runs, as shown in Table I, the grafting yield was selected as target.

It can be seen from Tables I and II that, when the monomer ratio is 1/0, the immersing time is 15 min, the initiator is 2.5%, the crosslinking agent is 30%, and the interface agent is 60%, the grafting yield will be optimal. However, we desire to get cografing onto PP fabrics with the two monomers other than only AA.

As shown in Table II, the initiator content is the second most important factor on grafting yield, but it has been widely

Table I. Test Schedule for Grafting of Binary Monomers of GMA and AA

Volume ratio (GMA/AA)	Immersing time (min)	Initiator (W_i/W_o , %)	Crosslinker (W_c/W_m , %)	Interfacial agent (W_f/W_m , %)	Gy (%)
0/1	10	0.5	10	20	228 ± 8
0/1	15	1.5	40	80	563 ± 20
0/1	20	3.5	20	40	471 ± 20
0/1	25	2.5	30	60	553 ± 20
3/7	10	3.5	20	60	396 ± 16
3/7	15	2.5	30	40	400 ± 15
3/7	20	0.5	10	80	286 ± 10
3/7	25	1.5	40	20	333 ± 13
7/3	10	2.5	40	40	321 ± 13
7/3	15	3.5	10	60	305 ± 13
7/3	20	1.5	30	20	311 ± 13
7/3	25	0.5	20	80	263 ± 10
1/0	10	1.5	30	80	338 ± 13
1/0	15	0.5	20	20	196 ± 8
1/0	20	2.5	40	60	316 ± 13
1/0	25	3.5	10	40	224 ± 9

Note: W_i is mass of initiator, W_o is volume of total grafting solution (the density of BPO was assumed one), W_c is volume of crosslinking agent, W_f is volume of rapeseed oil, W_m is volume of two monomers, and Gy is grafting yield.

Table II. Sum of Grafting Yield at Each Level Based on Five Factors (Gy)

Factors	Monomer ratio (GMA/AA)	Immersing time (min)	Initiator (W_i/W_o , %)	Crosslinker (W_c/W_m , %)	Interfacial agent (W_f/W_m , %)
Sum of level 1	1815 ± 68	1283 ± 50	973 ± 36	1043 ± 40	1068 ± 42
Sum of level 2	1415 ± 54	1464 ± 56	1545 ± 59	1533 ± 59	1450 ± 53
Sum of level 3	1200 ± 49	1384 ± 56	1396 ± 58	1326 ± 54	1416 ± 57
Sum of level 4	1074 ± 43	1373 ± 52	1590 ± 61	1602 ± 61	1570 ± 62
Polar difference	741 ± 111	181 ± 102	617 ± 97	559 ± 101	502 ± 104

investigated in many earlier works. Immersion time at longer than 15 min showed little influence on grafting yields. It is really surprising to find out that interfacial compatilizer and crosslinker are very important factors, but they have not been paid much attention so far.

Although the reactivity of GMA is weaker than that of AA, and the grafting yield of GMA seems to be increased due to its good solubility in rapeseed oil. The function of compatilizer, rapeseed oil, may include several aspects: (1) promoting monomer uptake onto PP fiber surface owing to swelling, especially for only AA and GMA grafting, as clearly shown in Table I, namely the higher content oily compatilizer, and the better grafting yield; (2) serving as diluent for monomers and initiator other than solvent; (3) acting as a plasticizer and avoiding the fragile peeling off from the fabrics; (4) dissipation of grafting polymerization heat and getting grafting process smooth. In addition, the reactivity of trace amount of carbon-carbon double bonds in the compatilizer was not taken into consideration. When AA accounts for a large

fraction, that is, 3/7, both overload and underload of compatilizer were not a good choice. The compatilizer is more effective to enhance the grafting yield of AA than that of GMA, indicating that a less oil compatilizer leads to a higher grafting at GMA dominant composition.

In this work, crosslinking agent is attempted to boost grafting and to alter the scattering grafting granule impression as mentioned. Crosslinker reacts with monomers and forms crosslinked polymers on PP fiber surface, and the crosslinking structure can imbibe more monomers to graft polymerization. However, the highly crosslinked grafting polymers have little wettability and thus peel off from fiber surface.

Since grafting copolymerization just took place in oven as well as in a sandwich mode, there should be oxygen both in grafting solution and inside fabric. In this case, BPO is partially used to counteract the inhibition of oxygen and makes up insufficient PP macromolecular radicals produced during glow-like discharging treatment, and thus BPO is expected to improve grafting yield.

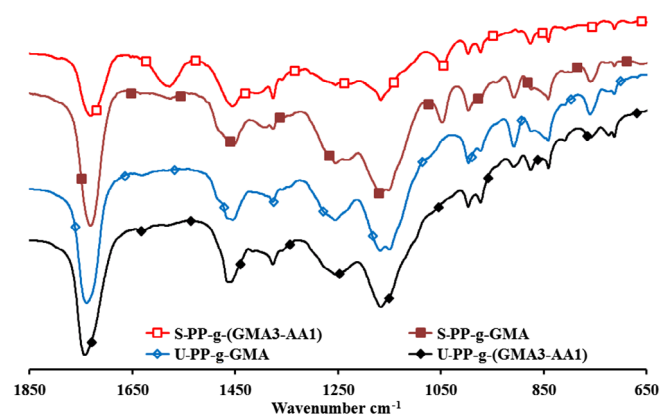


Figure 2. Infrared spectra of PP-g-(GMA-AA) and S-PP-g-(GMA-AA) (note: $W_i/W_0 = 2.5\%$, $W_c/W_m = 30\%$, $W_f/W_m = 60\%$, S-PP-g-(GMA-AA) implies the sulfonated polypropylene fabric-grafted with poly(glycidyl methacrylate) and poly(acrylic acid), PP-g-(GMA3-AA1) means the weight feed ratio of GMA and AA at 3/1.) [Color figure can be viewed at wileyonlinelibrary.com]

However, too many primary free radicals, namely overload initiator, will tend to promote homopolymerization. In fact, this grafting copolymerization more or less resembles the curing of unsaturated ester resin, in which curing reaction happened also in air and no measures was taken to purge oxygen in the resin.³⁰

As shown in Figure 2, two curves of the unsulfonated fabrics are listed just for close contrast. The new peak at 1560 cm^{-1} is attributed to asymmetric stretch of carboxylic group from AA, and it is caused by neutralizing at alkaline condition. Similarly, the new peak at 1043 cm^{-1} is assigned to stretch of sulfonate due to epoxy sulfonation. The peak at 908 cm^{-1} still originates from the absorption of epoxy group from GMA moiety. After sulfonation, the peak 908 cm^{-1} for the sample S-PP-g-(GMA3-AA1) completely disappears, while that of S-PP-g-GMA only slightly reduces. If the peaks at 2930 and 2850 cm^{-1} from stretch of methyl and methylene groups are appointed as internal reference, the peak area of 908 cm^{-1} of S-PP-g-GMA is decreased by 30%, implying that the sulfonation yield is approximately 30%. Based on the elemental analysis of SEM-EDS results listed in Table III, sulfonation reaches 36.1% for S-PP-g-GMA. Therefore, AA ingredient really accelerates sulfonation of epoxy group.

However, when AA is dosaged, that is, the volume ratio of GMA/AA = 7/3, only small fraction of GMA can be incorporated into grafting chains since the reactivity of GMA is weaker than that of AA. In fact, the epoxy groups of the samples PP-g-(GMA3-AA2), PP-g-(GMA7-AA3), and PP-g-(GMA3-AA3) could not be found clearly with IR spectroscopy, and no sulfur element after sulfonation operation would be detected with SEM-EDS for the three samples as well. Actually, in the light of results in Table III and further calculation, the molar ratio of GMA to

Table III. Elemental Composition of Sulfonated and Grafted Fabrics at Different Conditions

Samples	Element symbol	Atomic number (%)	Error (%)	Sulfonation degree (%)	Molar yield (%)
S-PP-g-GMA, sulfonated at alkaline conditions (optimal run)	C	73.6	1.6	15.1	97
	O	25.3	0.9		
	S	1.1	0.2		
S-PP-g-GMA, sulfonated at neutral conditions (optimal run)	C	70.8	1.6	4.3	144
	O	28.8	1.0		
	S	0.4	0.2		
S-PP-g-GMA, sulfonated at alkaline conditions (7/3/2 run)	C	82.8	1.6	36.1	21
	O	15.8	1.0		
	S	1.4	0.2		
S-PP-g-(GMA3-AA1), sulfonated at alkaline conditions (7/3/2 run)	C	85.6	1.5	100	25
	O	13.7	1.0		
	S	0.7	0.2		
S-PP-g-(GMA3-AA2), sulfonated at alkaline conditions (7/3/2 run)	C	80.0	1.5	0	-
	O	20.0	1.0		
S-PP-g-(St3-AA3) (no SDBS)	C	83.8	1.5	100	54
	O	14.7	1.0		
	S	1.5	0.2		
S-PP-g-(St3-AA2) (no SDBS)	C	90.6	1.4	0	-
	O	9.4	1.0		
S-PP-g-(St3-AA1) (no SDBS)	C	93.1	1.3	0	-
	O	6.9	1.0		

Note: S-PP-g-(St3-AA1) implies the sulfonated polypropylene fabric-grafted with polystyrene and poly(acrylic acid), the monomer feed mass ratio of St and AA is at 3/1, no SDBS indicates no addition of sodium dodecyl benzene sulfonate.

AA in the grafting polymers becomes 0.15/1 at the feed molar ratio 1.44/1.0. Conclusively, the dosaging of AA significantly reduces copolymerization ability of GMA, only small amount of GMA is incorporated into grafting chains.

The estimation of reaction degree of epoxy group is made by assuming that all oxygen atoms come from GMA prior to sulfonation, and the sulfonation degrees of epoxy group are approximately 15.1 and 4.3% for PP-g-GMA at alkaline and neutral conditions, respectively. It means that alkalinity promotes sulfonation. This finding is consistent with the conclusion of Ueki *et al.*³¹ In addition, the calculated grafting yields are also listed in Table III, which was derived from the results of element contents. Ninety seven percentage of molar grafting yield is corresponding to 329% of weight grafting, and this sounds rational. Then, while the two samples S-PP-g-GMA and S-PP-g-(GMA3-AA1) are compared, the increase of molar grafting of GMA is quite reliable from 21 to 25%. Thus, this implies the convincing of the assumption of complete sulfonation of PP-g-(GMA3-AA1).

Contrary to the finding of PP-g-(GMA-AA), PP-g-(St-AA) fabrics have not been sulfonated at the low feed of AA. As shown in Figure 3, the peaks appeared at 1604, 1499, 1452, and 700 cm^{-1} are attributed to grafted polystyrene fractions. The twin peaks at 970 and 997 cm^{-1} originates PP matrix as mentioned above. The peak at 1710 cm^{-1} would come from AA moiety. Therefore, the polystyrene and polyacrylic acid segments are certainly grafted onto PP fabric by thermal polymerization. However, there appears new shoulder peak at 1735 cm^{-1} for the samples PP-g-(St3-AA3) and PP-g-(St3-AA2), and this will be explained later.

After sulfonation, great change areas in IR spectra are appeared in the ranges of 1450–1150 cm^{-1} and 1750–1650 cm^{-1} and imply a bit complicated process. The intensified peaks at the areas of 1450–1150 cm^{-1} symbolize sulfonation, the stretch vibration of sulfonyl group. The relative change of twin peak depths at 1735 and 1710 cm^{-1} is very strange, since sulfonation seems nothing related to them. In addition, the peak reductions at 700 and 1499 cm^{-1} evidence the weakening of polystyrene moiety. The high frequency side of the peak 700 cm^{-1} appears new shoulder peak after sulfonation, which coincides with the results of

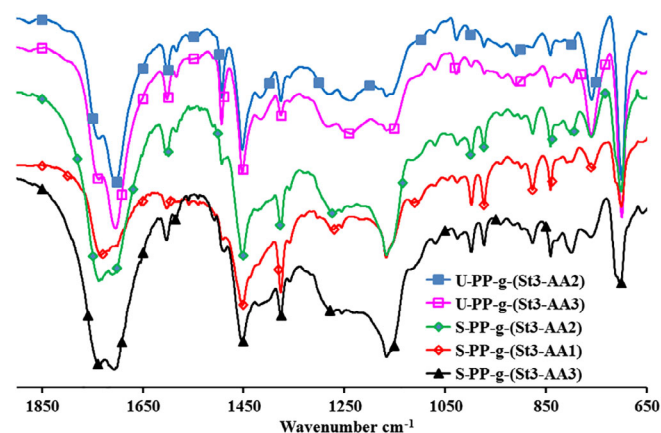


Figure 3. Infrared spectra of PP-g-(St-AA) and S-PP-g-(St-AA) (no oil) (note: no oil implies none of the oil compatilizer). [Color figure can be viewed at wileyonlinelibrary.com]

Phadnis' *et al.*^{32–34} St and AA monomers are cogenerated and thus may generate alternating polymer chains onto PP fabrics. In this case, the neighboring propyl acid reacts with benzene ring to form cyclic acylation under catalysis of concentrated sulfuric acid, with ring compound containing carbonyl group being produced. The formation of carbonyl group prior to sulfonation can be explained by acylation under heating,³⁴ since the thermal grafting took place even at 130 °C for 20 min, only acylation reaction happens more violently by catalysis of sulfuric acid than heating. Acylation alters monosubstituted to bisubstituted benzenes or heterogeneous ring compounds, as indicated by the peak change of 1499 and 700 cm^{-1} .

Although the molar ratios of St to AA in this study were 1.70/1.0, 1.0/1.18, and 1.0/1.77, a bit difference from the feed ratio of St to AA at 1.08/1.0 in the literature.^{32,33} Wong *et al.* found that the random copolymer of St and AA had one glass transition temperature, and the temperature was even higher than those of their block copolymer.³⁵ This implies the cyclization of propyl acid with the benzene ring of St even happens in random copolymer. Conclusively, even nonideal alternative copolymer of St and AA in this study can also proceed with acylation reaction only in an alternative segment. That's why PP-g-(St3-AA3) and PP-g-(St3-AA2) give shoulder peak at 1735 cm^{-1} even prior to sulfonation. Acylation reaction will accompany the loss of carboxylic group and thus lower the hydrophilicity of grafting polymers. Consequently sulfonation of PP-g-(St-AA) is diminished, especially for the samples PP-g-(St3-AA2) and PP-g-(St3-AA1).

As shown in Table III, the molar grafting yield of St moiety of S-PP-g-(St3-AA3) reaches 54%, corresponding to weight grafting at 122%, which is comparable with experimental value, 198% of the weight grafting of PP-g-(st3-AA3) as referred to Table IV. Therefore, the assumption of complete sulfonation of St moiety of PP-g-(St3-AA3) is convincing, and the inclusion of hydrophilic fraction improves sulfonation.

Morphology of Dual Monomers Grafting Fabrics

As shown in Figure 4(a), PP-g-GMA looks like a corrugated pipe, which is interpreted as dissipation token of grafting stress of poly(glycidyl methacrylate) layer owing to low glass transition temperature. However, the surface tension, polarity, and glass transition temperature of grafted poly(acrylic acid) are normally larger than those of poly(glycidyl methacrylate), and this decides granular forms on the highly crystalline PP fiber surface, as shown in 4(b). Surprisingly, when the monomer ratio is at 7/3, the morphology of PP-g-(GMA-AA) is totally different as shown in Figure 4(c). The grafting polymer is wrapped tightly onto the fiber surface. Therefore, the grafting of dual monomer seems to

Table IV. Adsorption and Electrochemical Desorption of S-PP-g-(St3-AA2)

Repetition times	1	2	3
Ac (mg/g)	45.6 ± 3.3	43.5 ± 3.5	40.7 ± 2.1
E _d (%)	56.7 ± 4.3	43.7 ± 5.2	49.6 ± 4.2
Ce (%)	48.2 ± 6.4	33.3 ± 5.5	36.7 ± 5.2

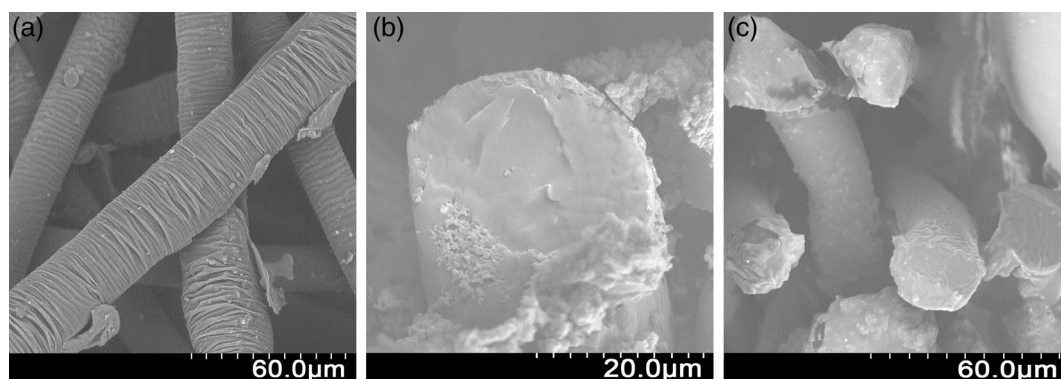


Figure 4. SEM images of fiber surface morphology of grafted PP fabrics (a: PP-g-GMA, Gy = 40%; b: PP-g-AA, Gy = 100%; c: PP-g-(GMA7-AA3) Gy = 70%; 7/3/2 run).

improve wettability and film-formation of poly(acrylic acid) segment on fiber surface. This morphology change may imply crosslinking reaction between epoxy and carboxyl groups during grafting to some extent, which is consistent with previous report by Orr *et al.*³⁶

Figure 5 displays SEM images of the nonwoven surfaces grafted with GMA and AA before and after sulfonation. At the high dosage of compatilizer, together with the high grafting yield, the grafting morphology switches from the corrugation film around fibers into granules as shown in Figure 5(a). In this case, the compatilizer seems to act as emulsifier to some extent. Nevertheless, on the fiber surface still exists a skin layer, similar to that composed of smooth layer and granules as shown in the right and down part of Figure 5(a). Thermal polymerization prefers more or less on the outside of fabric in our experimental setup, indicating that grafting copolymers may pile on the fabric surface other than that uniformly distributed on exterior and interior of the fabric, as shown in Figure 5(b). After sulfonation, some granules fall apart, and some sulfonated grafting polymers still adhere on the fiber surface, as indicated in Figure 5(c). Although the granular dropping may ruin grafting yield, the large specific surface areas are also good for functionalization. The skins filled with particles around fiber surface are still wrapped after sulfonation, as indicated in Figure 5(d). Although the SEM sample preparation seems to shake off sticking grafting polymers, the skin layer composed of smooth film and particles is tough enough even after being suffered by sulfonating.

Figure 6 gives the SEM images of fiber and fabric surfaces of PP-g-(St3-AA1) and S-PP-g-(St3-AA1), the grafting copolymers are

presented in a smooth layer filled with particles prior to sulfonation as shown in Figure 6(a). Since the feed ratio of St to AA was at 3/1, or the molar ratio 1.70/1.0, it means that this monomer proportion cannot generate the strict alternative copolymer, the disproportion would produce both single component and alternative copolymer chains, and this gives rise to microphase segregation structure. Nevertheless, the granule feature seems disappeared after sulfonation, and this can be assigned to hydrophilicity of the sulfonated grafting polymers. The grafting copolymers are uniformly wrapped on the fiber surface, and the PP fiber still keeps good tenacity as indicated in Figure 6(c). However, the PP fiber appears fragile after sulfonation, and this may be caused by oxidation of sulfuric acid as demonstrated in Figure 6(d). It is worth nothing that, although there is particle filling layer wrapped on fiber, no isolated or densely packed particles appear in the absence of oil and SDBS.

As shown in Figure 7(a), the fiber feature for high grafting yield almost totally disappears and the grafted copolymers cover on the surface nonuniformly, which may be caused by grafting operation. The voids on the surface may originate to the entrapped air, while being covered with Teflon sheets. The grapery particles would be assigned to microdomain segregation in Figure 7(b). Figure 7(c) presents the microstructure of the sulfonated and grafted fabric, and much loosely packed particle structure appears, noting that the particle-filling film morphology is changed into particle aggregates. The reason is that the oxidation of concentrated sulfuric acid disconnects the links between microdomains and makes the segregated structure into loosely packed particles. Sulfonation and oxidation of grafting polymers happen

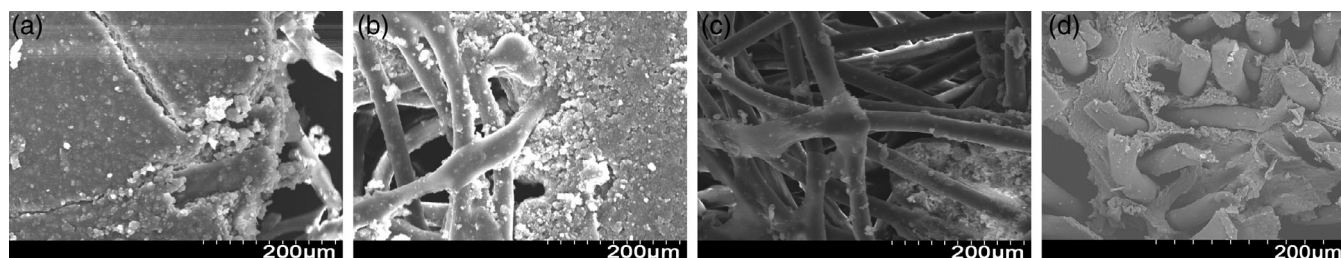


Figure 5. SEM images of fiber and fabric surfaces of grafted PP (Note: a: PP-g-GMA, Gy = 260%; b: PP -g-(GMA3-AA2), Gy = 266%; c: S-PP-g-GMA, Gy = 260%; d: section of PP-g-(GMA3-AA2), Gy = 266% (optimal run)).

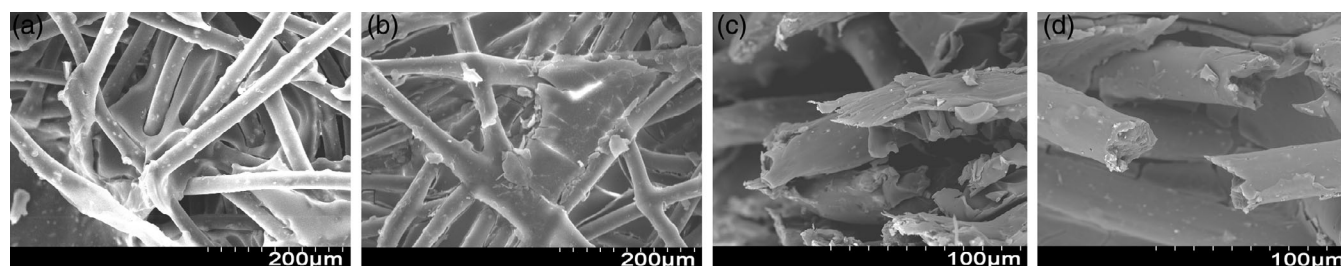


Figure 6. SEM images of fiber and fabric surfaces of PP-g-(St3-AA1) and S-PP-g-(St3-AA1) at Gy = 63% (Note: a: unsulfonated; b: sulfonated; c: unsulfonated; d: section of sulfonated, no SDBS).

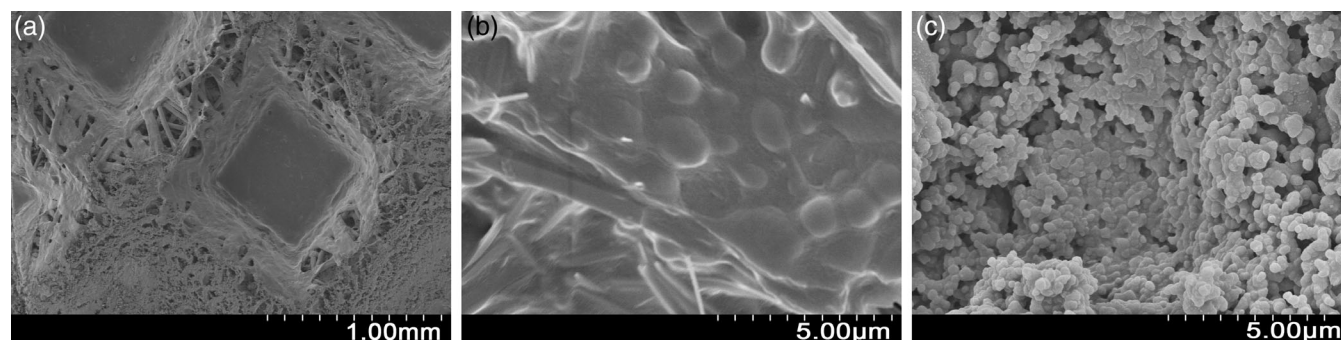


Figure 7. SEM images of fabric surface of PP-g-(St3-AA2) and S-PP-g-(St3-AA2) at Gy = 188% (Note: a: unsulfonated; b: unsulfonated; c: sulfonated, oil, and SDBS).

simultaneously. Since single St grafting polymers cannot be sulfonated in our experimental setup, it is the incorporation of AA that accelerates sulfonation of polystyrene fraction. The sulfonated particle aggregates may closely relate to oil compatilizer simply due to emulsification, and this also intensifies sulfonation by increasing reactive surface.

Electrochemical Desorption and Repetitive Utilization

Considering that the ion exchange fabrics possess a few advantages in adsorption and separation over ion exchange resin, such as, easy to separate from aqueous solution, convenient to regenerate without acid, we performed electrochemical desorption experiment for as-modified fabrics. As shown in Figure 8, the higher the content of acrylic acid is, the larger the adsorption capacity is. Actually, a higher metal uptake carboxymethyl cellulose, a better ion selective nut husk adsorbent, and a faster alkali absorption battery separator membrane all have evidenced powerful combination of carboxylic and sulfonic groups.^{3,5,6} Our earlier thesis work showed that the adsorption capacity of sulfonic group towards copper (II) ion was much higher than that of carboxylic one. Consistently Morales *et al.* found that both poly(2-acrylamidoglycolic acid-co-2-acrylamide-2-methyl-1-propane sulfonic acid) and poly(2-acrylamidoglycolic acid-co-4-St sodium sulfonate) resins showed faster metal ion removals than that of Amberlite IRP-64 resin, a only carboxylic group containing resin.³⁷

Assuming that the grafting polymer ingredients are same as the feed ones, all styrene moieties are sulfonated, one sulfuric and two carboxyl groups uptake one copper ion, respectively, finally the calculated adsorption capacity is listed as $A_{c,cal1}$. If ideal

alternative copolymers are indeed produced, and cyclization compounds are thus realized completely, the capacity is indicated as $A_{c,cal2}$. By comparing the two calculations with real experimental one, it will be found out that the copolymer and feed compositions cannot be same, not all styrenes will be sulfonated, and ideal copolymer chain and complete cyclization will not happen as well.

If the sulfonation yield of polystyrene fraction was the same, the increase of AA meant the decrease of St fraction, and then total adsorption amount of the sulfonated fabric should be reduced. This is contrary to real experimental case. Therefore, it is deduced that AA remarkably improves sulfonation of polystyrene segment and thus increases adsorption capacity, and only partial grafting copolymer chains are transformed into cyclization ones.

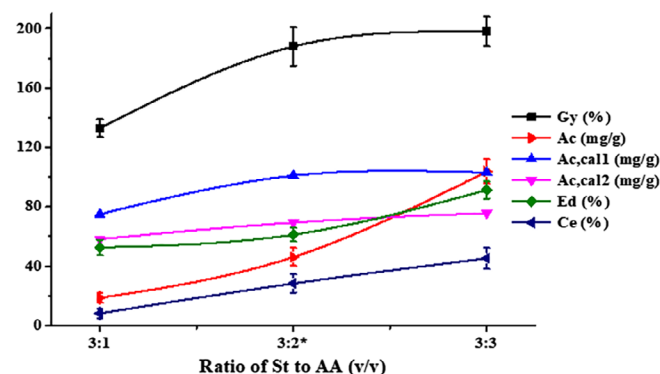


Figure 8. Adsorption and electrochemical desorption of S-PP-g-(St-AA). [Color figure can be viewed at wileyonlinelibrary.com]

Note that the sample 3/2 recipe was grafted in the presence of oil compatilizer. Ac is ion adsorption capacity. Ac, cal_1 is calculated with assumptions that the grafting polymer ingredients are same as the feed ones, all styrene moieties are sulfonated, one sulfuric and two carboxyl groups uptake one copper ion, respectively. Ac, cal_2 is calculated with assumptions that ideal alternative copolymers are indeed produced, the cyclization compounds are realized and sulfonated completely. E_d is the electrochemical desorption rate, which is defined as the weight ratio of the copper ions weight gain in solution to the copper ions weight adsorbed on the fabric. C_e is the electrochemical desorption current efficiency, which is calculated based on the ratio of the molar electric charge of copper ions desorbed to the total electric charge transfer in solution.

Also as shown in Figure 8, desorption rate and current efficiency are increased with the increase of AA content. Under DC voltage, the proton produced on the anode directly makes ion exchange with copper ions adsorbed on fabric, and then drives copper ions into electrolytic cell. The accumulation of copper ions in solution improves the electrolytic solution conductivity, and in turn the voltage drop in the electrode regions is increased, thus making that a more proton is generated, resulting higher electrochemical desorption rate and current efficiency.

Table IV gives the adsorption and electrochemical desorption of S-PP-g-(St3-AA2) after three repetitions. The adsorption capacity is gradually decreased with repetition, since more or less loosely packed particles would be lost especially at the initial stage. The electrochemical desorption rate and desorption current efficiency tend small. The slight undulation of desorption may relate microstructural self-adjustment of sulfonated polymers under brine and electric environments. Figure 9 demonstrates SEM images of the fabric and fiber surfaces after all three repetitions. The scattered particles on surface are lost, and the tiny sparkles on fiber are crystalline copper salts. Conclusively, compatilizer improves the wettability for grafting, but its dosage should be thought over and optimized indeed.

Table V displays the effective of the copper removal with S-PP-g-GMA, indicating that the sulfonic group almost instantly removes most of the copper (II) ion from simulated wastewater within 20 min. Although the sulfonation degrees of S-PP-g-GMA were different, the removals of copper ion seem quite same. This may be attributed to overloaded adsorbents. The copper removal

Table V. Copper (II) Ions Removal with S-PP-g-GMA (7/3/2 run)

S-PP-g-GMA	Removal (%)
1	98.7 ± 3.4
2	98.7 ± 3.2
3	98.8 ± 4.4
4	97.1 ± 2.9
5	95.7 ± 3.1

Note: the samples 1–3 were sulfonated in the alkaline conditions, while the samples 4–5 were sulfonated in neutral condition. Initial copper conc. was at 57.95 ± 2.5 mg/L.

results confirm the sulfonation of epoxy group successfully at mild condition other than corrosive and oxidative condition, although the reactivity of GMA along seems rather low in comparison with that of dual monomer-grafted fabric.

CONCLUSIONS

The grafting copolymerization and successive sulfonation were conducted successfully for PP nonwoven fabric. Both rapeseed oil and PETA promote grafting, but the over-dosage of the compatilizer unexpectedly contributes its side-effect emulsification and intensifies the microphase segregation of grafting polymer themselves. The granular grafting polymers exhibit superior functionalization but will deteriorate durable utilization. The grafting copolymerization of dual monomers, GMA and AA, St and AA, certainly promote sulfonation of epoxy group and benzene ring respectively, whereas the overload of AA suppresses copolymerization of GMA and causes acylation reaction of carboxyl group with neighboring benzene, and thus spoil sulfonation to some extent.

For the sample PP-g-(GMA-AA) preparation, the components of grafting solution include monomers (GMA7/AA3): PETA: oil = 7:3:2, and thermally grafting copolymerization onto PP fabric occurs at 120°C for 40 min. The ingredients of sulfonation solution are NaOH, Na₂SO₃, CH₃CH(OH)CH₃, and H₂O at the corresponding ratio of 0.2 g/20 g/30 mL/150 mL, and sulfonation of the grafted fabric happens at 80°C for 5 hours. For the sample PP-g-(St-AA) preparation, the components of grafting solutions include 40 mL St, 40 mL AA, 10 mL DVB, and 2.5 g BPO, and

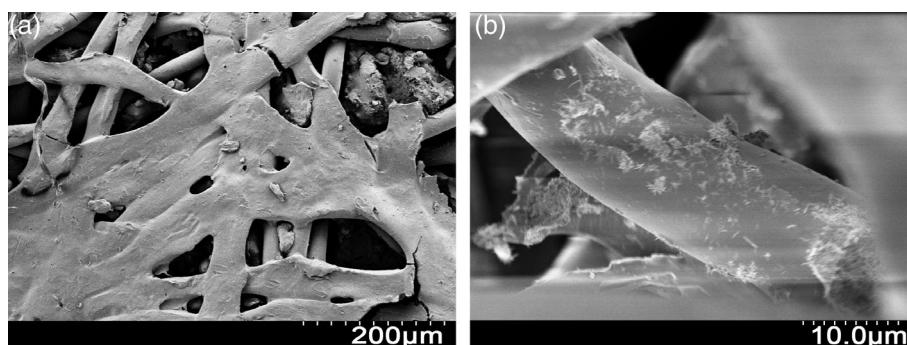


Figure 9. SEM images of fiber and fabric surface of S-PP-g-(St3-AA2) after adsorption and electrochemical desorption.

thermally grafting copolymerization takes place in an oven in three periods stepwise, 70°C 30 min, 100°C 20 min, and 130°C 20 min. The grafted fabric is sulfonated by immersing in 60 and 98% H₂SO₄ at ambient temperature for 1 h, respectively.

Both S-PP-g-(st-AA) and S-PP-g-(GMA-AA) showed good copper (II) uptake and fairly good reutilization. The sulfonated PP fabric, especially its sheet-like structure, can act as ion exchange fabrics and membranes for separation and purification of heavy metal ions wastewaters in the processes of membrane separation, electrodialysis, and electrodeionization.

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